The University of Tokyo
Graduate School of Science
Department of Physics — Master's Program Entrance Examination

Mathematics

August 20, 2018 (Monday)
9:30 AM – 11:00 AM

Instructions

1. Do not open this question booklet until the signal to start the exam is given.
2. Use only a black pencil (or black mechanical pencil) to write your answers.
3. There are two questions in total. Answer **all** of them.
4. Two answer sheets are provided, one per question. Ensure both sheets are received.
5. Fill in your examinee number and name in the designated space on the question booklet.
6. Fill in the subject (Mathematics), examinee number, name, and question number in the designated space on each answer sheet.
7. Use a separate answer sheet for each question.
8. Each answer sheet has a perforated edge and may be detached. If using the reverse side, do not write above the perforation line.
9. Do not write anything unrelated to your answers—such as letters, marks, or symbols—on the answer sheets.
10. Even if you cannot answer a question, submit the answer sheet with the subject (Mathematics), examinee number, name, and question number filled in.
11. Do not use the answer sheets as scratch paper. Scratch paper will be provided separately.

H31(2019)

Problem 1

Answer the following questions. Let X, Y be square matrices. For X, let $\mathrm{tr}X$ denote the trace, and $\det X$ the determinant. Define $[X, Y] = XY - YX$, and

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

Let $X^\dagger$ denote the Hermitian conjugate (adjoint) of X, and similarly for vectors. Let I be the identity matrix and i the imaginary unit.

1. Let A and B be Hermitian matrices ($A^\dagger = A$, $B^\dagger = B$), and traceless ($\mathrm{tr}A = \mathrm{tr}B = 0$). Show the following (i)–(iii). You may use that a Hermitian matrix A can be diagonalized as $A = U^\dagger D U$ by a unitary matrix U ($V^\dagger V = I$), and that for invertible matrices X, Y, $\det(XY) = \det X \cdot \det Y$.

   (a) A can be diagonalized by a unitary matrix U into a traceless diagonal matrix.

   (b) $U^\dagger U = I$ is a unitary matrix.

   (c) $U = e^{iA}$ satisfies $\det U = 1$.

   Any traceless 3×3 Hermitian matrix T can be expressed using 8 real parameters $\alpha_1, \ldots, \alpha_8$ and 8 traceless 3×3 Hermitian matrices $T_1, T_2, \ldots, T_8$ as:

$$T = \begin{pmatrix} \alpha_7 & \alpha_1 - i\alpha_2 & \alpha_3 - i\alpha_4 \\ \alpha_1 + i\alpha_2 & \alpha_8 & \alpha_5 - i\alpha_6 \\ \alpha_3 + i\alpha_4 & \alpha_5 + i\alpha_6 & -\alpha_7 - \alpha_8 \end{pmatrix} = \sum_{a=1}^{8} \alpha_a T_a.$$

2. Write down the explicit forms of the matrices T_1, T_2, and T_7.

3. From 1(i), define real constants A_{abc} $(a, b, c = 1, \ldots, 8)$ such that

$$[T_a, T_b] = i \sum_{c=1}^{8} A_{abc} T_c.$$

   Find A_{112} and A_{566}.

4. Define the matrix $U = e^{iT}$ where

$$T = \sum_{a=1}^{8} \alpha_a T_a.$$

   Consider complex 3-component vectors $x = (x_1, x_2, x_3)^T$, $y = (y_1, y_2, y_3)^T$, and their transformations $x' = Ux$, $y' = Uy$.

   (a) Let $G_a = x^\dagger T_a y$, $G'_a = x^\dagger T_a y'$. Show that to first order in α,

$$G'_a = G_a + i \sum_{b,c} \alpha_b G_c A_{bca} + \mathcal{O}(\alpha^2),$$

   where $H_{abc} = G_c A_{bca}$.

   (b) Let $z_i = \epsilon_{ijk} x_j y_k$, $z'_i = \epsilon_{ijk} x'_j y'_k$ and define $z = (z_1, z_2, z_3)^T$, $z' = (z'_1, z'_2, z'_3)^T$. Show that

$$z' = (I + iH)Mz + \mathcal{O}(\alpha^2),$$

   where H is an antisymmetric 3×3 matrix and M satisfies the identity

$$\sum_{c=1}^{8} A_{abc} A_{dec} = \delta_{ad}\delta_{be} - \delta_{ae}\delta_{bd}.$$

Problem 2

Consider the following operator acting on functions defined on the interval $x \in [0, \pi]$:

$$H_\alpha = -\frac{d^2}{dx^2} - m_\alpha^2,$$

where m_α is a positive real number and $\alpha = 1, 2$. Let $\lambda_n^{(\alpha)}$ be the n-th smallest eigenvalue of H_α, and denote the corresponding eigenfunction by $f_n^{(\alpha)}(x)$ ($n = 1, 2, 3, \dots$), such that

$$H_\alpha f_n^{(\alpha)}(x) = \lambda_n^{(\alpha)} f_n^{(\alpha)}(x).$$

The eigenfunctions $f_n^{(\alpha)}(x)$ satisfy the following boundary conditions:

$$\left.\frac{df_n^{(\alpha)}(x)}{dx}\right|_{x=0} = 0, \quad f_n^{(\alpha)}(\pi) = 0.$$

We assume $f_n^{(\alpha)}(x) \neq 0$ (i.e., nontrivial solutions).

1. Find $\lambda_n^{(\alpha)}$ and $f_n^{(\alpha)}(x)$. You do not need to normalize the eigenfunctions.
2. Assume $m_\alpha < \frac{1}{2}$ and define

$$D = \prod_{n=1}^{\infty} \frac{\lambda_n^{(1)}}{\lambda_n^{(2)}}.$$

   Introduce the analytic function $Z(s)$ defined by

$$Z(s) = \sum_{n=1}^{\infty} \left[\left(\lambda_n^{(2)}\right)^{-s} - \left(\lambda_n^{(1)}\right)^{-s} \right].$$

3. Express D using $Z'(0)$.
4. Let $\lambda > -m_2^2$ be a real number. Find the solution $g_\alpha(x; \lambda)$ to the differential equation and boundary conditions:

$$H_\alpha g_\alpha(x;\lambda) = \lambda g_\alpha(x;\lambda), \quad g_\alpha(0;\lambda) = 1, \quad \left.\frac{\partial g_\alpha(x;\lambda)}{\partial x}\right|_{x=0} = 0.$$

5. Use the value of $g_\alpha(\pi; \lambda)$ obtained from the above to define

$$\hat{g}_\alpha(\lambda) = g_\alpha(\pi; \lambda).$$

Assuming $\hat{g}_\alpha(\lambda)$ is an analytic function, it has isolated zeros at $\lambda = \lambda_n^{(\alpha)}$ $(n = 1, 2, 3, \dots)$, with expansion near $\lambda = \lambda_n^{(\alpha)}$:

$$\hat{g}_\alpha(\lambda) = (\lambda - \lambda_n^{(\alpha)}) z_n^{(\alpha)} + \mathcal{O}\left((\lambda - \lambda_n^{(\alpha)})^2\right).$$

Show that

$$Z(s) = \int_C \lambda^{-s} \left[\frac{d}{d\lambda} \log \hat{g}_2(\lambda) - \frac{d}{d\lambda} \log \hat{g}_1(\lambda)\right] d\lambda,$$

where the contour C encircles all possible eigenvalues $\lambda_n^{(\alpha)}$ in the complex plane. The contour must avoid the branch cut of λ^{-s} and ensure convergence.

6. Use the contours C_{in} and C_{out} as shown in the figures to deform C appropriately and evaluate $Z(s)$.

7. Compute D explicitly when $m_1 = m_2$ and express your answer.

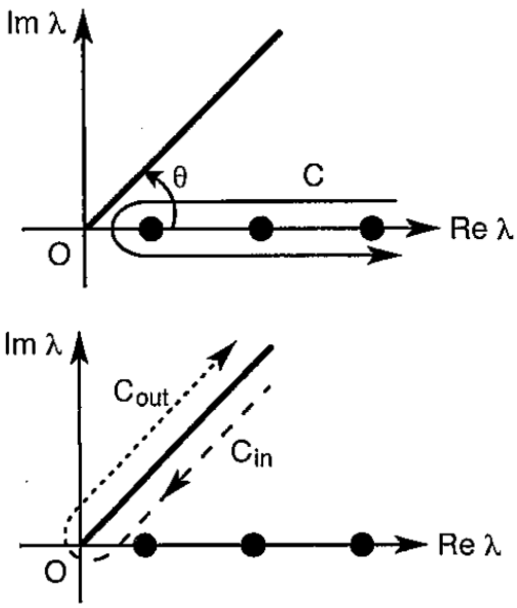

Figure 1: Left: Contour C (solid), C_{in} and C_{out} (dashed). Dots are $\lambda_n^{(\alpha)}$. Large real eigenvalues accumulate toward the branch cut.